

Regression Coefficients — Line-by-Line Calculation

This PDF explains how Excel calculates Coefficients, Standard Error, t Stat, P-value, and the 95% confidence intervals.

1. Regression equation

The regression equation is:

$$\hat{Y} = a + bX = 74.55655351 + 3.018572081X$$

Therefore:

$$\text{Intercept (a)} = 74.55655351$$

$$\text{Advertising Spend coefficient / slope (b)} = 3.018572081$$

2. Important values used in every calculation

$$n = 15 \text{ observations}$$

$$k = 1 \text{ independent variable}$$

$$\text{Residual degrees of freedom} = n - k - 1 = 15 - 1 - 1 = 13$$

$$\text{Mean } X = 27.466667$$

$$\sum(X - \bar{X})^2 = 1751.733333$$

$$\text{Residual SS (SSE)} = 8.329121632$$

$$\text{MSE} = \text{SSE} / \text{df} = 8.329121632 / 13 = 0.640701664$$

3. Coefficients column

Intercept coefficient:

$$\text{Formula: } a = \bar{Y} - b\bar{X}$$

$$= 157.466667 - (3.018572081 \times 27.466667)$$

$$= 74.55655351$$

$$\text{Intercept} = 74.55655351$$

Advertising Spend coefficient:

$$\text{Formula: } b = \sum(X - \bar{X})(Y - \bar{Y}) / \sum(X - \bar{X})^2$$

$$= 5287.733333 / 1751.733333$$

$$= 3.018572081$$

$$\text{Advertising Spend coefficient} = 3.018572081$$

4. Standard Error of the Advertising Spend coefficient

Formula:

$$\text{SE}(b) = \sqrt{[\text{MSE} / \sum(X - \bar{X})^2]}$$

$$= \sqrt{[0.640701664 / 1751.733333]}$$

$$= \sqrt{0.000365752967}$$

$$= 0.019124669$$

$$\text{Excel value: } 0.019124669$$

5. Standard Error of the Intercept

Formula:

$$SE(a) = \sqrt{[MSE \times (1/n + X^2 / \sum(X - \bar{X})^2)]}$$

$$= \sqrt{[0.640701664 \times (1/15 + (27.466667)^2 / 1751.733333)]}$$

$$= 0.564485593$$

Excel value: 0.564485593

6. t Stat for Intercept

The t statistic tells us how many standard errors the coefficient is away from zero.

Formula: $t = \text{Coefficient} / \text{Standard Error}$

Intercept $t = 74.55655351 / 0.564485593$

$= 132.0787535$

t Stat = 132.0787535

7. t Stat for Advertising Spend

Formula: $t = \text{Coefficient} / \text{Standard Error}$

$t = 3.018572081 / 0.019124669$

$= 157.8365654$

t Stat = 157.8365654

8. P-value

For a two-sided t-test, Excel calculates the probability of getting a t statistic at least this extreme when the true coefficient is zero.

Formula concept: $P\text{-value} = 2 \times P(T(df) \geq |t|)$

Degrees of freedom = 13

Intercept: $t = 132.0787535 \rightarrow P\text{-value} \approx 1.01033E-21$

Advertising Spend: $t = 157.8365654 \rightarrow P\text{-value} \approx 9.98120E-23$

Both p-values are extremely small, so both coefficients are statistically significant.

9. 95% Confidence Interval

The 95% confidence interval formula is:

$\text{Coefficient} \pm t\text{-critical} \times \text{Standard Error}$

For $df = 13$ and 95% confidence: $t\text{-critical} \approx 2.1603687$

10. Intercept 95% Confidence Interval

Lower 95% = $\text{Coefficient} - t\text{-critical} \times \text{SE}$

$= 74.55655351 - (2.1603687 \times 0.564485593)$

$= 73.33705653$

Upper 95% = $\text{Coefficient} + t\text{-critical} \times \text{SE}$

$= 74.55655351 + (2.1603687 \times 0.564485593)$

$= 75.77605049$

11. Advertising Spend 95% Confidence Interval

Lower 95% = $\text{Coefficient} - t\text{-critical} \times \text{SE}$

$= 3.018572081 - (2.1603687 \times 0.019124669)$

$= 2.977255745$

Upper 95% = $\text{Coefficient} + t\text{-critical} \times \text{SE}$

$= 3.018572081 + (2.1603687 \times 0.019124669)$

$= 3.059888417$

12. What Lower 95.0% and Upper 95.0% mean

Excel displays the same 95% confidence interval twice: Lower 95% / Upper 95% and Lower 95.0% / Upper 95.0%.

Therefore:

Lower 95% = Lower 95.0%

Upper 95% = Upper 95.0%

They are not additional calculations.

13. Final table

Term	Coefficient	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	74.55655351	0.564485593	132.0787535	1.01033E-21	73.33705653	75.77605049
Advertising Spend	3.018572081	0.019124669	157.8365654	9.98120E-23	2.977255745	3.059888417

14. Easy memory method

1. Coefficient = regression slope/intercept.
2. Standard Error = how much uncertainty is in the coefficient.
3. t Stat = Coefficient ÷ Standard Error.
4. P-value comes from the t distribution using df = 13.
5. 95% CI = Coefficient ± t-critical × Standard Error.
6. Lower 95% = Coefficient – margin of error.
7. Upper 95% = Coefficient + margin of error.
8. Lower 95.0% and Upper 95.0% are duplicates of Lower 95% and Upper 95% in this Excel output.

15. Interpretation in very simple words

The Advertising Spend coefficient is about 3.0186. This means that, according to the fitted linear model, increasing Advertising Spend by 1 unit is associated with an estimated increase of about 3.0186 Sales units.

The 95% confidence interval for the Advertising Spend coefficient is approximately 2.9773 to 3.0599. Since zero is not inside this interval, the coefficient is statistically significant at the 5% level.